

7. Data Mining is a process of analysing large datasets. With data mining tools and methods organizations can discover hidden patterns and relationships in their data. Data mining transforms raw data into practical knowledge. In business data mining helps identify trends, predict outcomes and optimize operations. ie Financial institutions it can greatly assist in fraud detection by seeing unusual transactions or patterns that indicate fraud.

In Healthcare it assists in using patient data to reduce appointment no shows, improve treatment plans, and enhance patient engagement.

8. Database Management System (DBMS)

Is a specialized software used for data storage, retrieval and manipulation. In order to efficiently manage and arrange data, it serves as a mediator between the database, applications, and user interfaces. The system offers a full range of tools to manage databases guaranteeing data integrity, security and consistency.

MySQL	PostgreSQL	MongoDB	ORACLE
Relational database syst.	Relational DBMS, built	Non-relational	Relational DBMS
Open source solution, Oracle	for complex data structure	Open source with	runs on Oracle &
Works well in Windows, Linux	Great scalability, JSON	commercial version	Strong tech support
MacOS, UNIX. Free installation	XML & HSTORE.	Fast, easy data op.	Large capacity
Simple syntax & mild complexity	Easily integrated third	Easy compatible	High cost
Compatible with cloud providers	party tools. Open source	with other models.	Hard learning curve
Scalability challenges.	community driven support.	Horizontal Scalable	Not Online OLTP
OLAP/OLTP Systems supported.	Lacks reporting and	solution. Lacks	data warehouse.
	auditing instruments.	data security.	